Data Warehousing & Data Mining - Exam Question Bank

Section A: Descriptive Questions (3 Marks Each)

Instructions: Answer broadly in 3-4 distinct *points or a comprehensive paragraph*.

1. Discuss data pre-processing. Mention any 1 objective.

Answer: Data pre-processing is the step in data mining where raw data is transformed into a clean and understandable format. Real-world data is often incomplete, inconsistent, and lacking in certain behaviors, and is likely to contain many errors.

- **Objective:** To improve the quality of the data (accuracy, completeness, consistency) so that the mining process is efficient and the results are reliable.

2. Differentiate between Data Cleaning & Data Integration.

Answer: | Data Cleaning | Data Integration | | :--- | :--- | | **Purpose:** Focuses on removing noise, handling missing values, and correcting inconsistencies in the data. | **Purpose:** Focuses on combining data from multiple sources (like databases, data cubes, or files) into a coherent store. | | **Key Task:** Filling in missing values, smoothing noisy data, identifying outliers. | **Key Task:** Handling schema integration (matching attributes from different entities) and resolving data value conflicts. | | **Example:** Replacing a `null` age value with the average age. | **Example:** Merging a `Sales_DB` and `Inventory_DB` using a common `Product_ID`. |

3. Define Explanatory Data Analysis (EDA).

Answer: Explanatory Data Analysis (often called Exploratory Data Analysis) is an approach to analyzing data sets to summarize their main characteristics, often using visual methods.

- It is used to understand the data's structure, detect outliers and anomalies, and test underlying assumptions.
- It helps determine the best statistical model to use for the data analysis.

4. Discuss any 2 common visualization techniques.

Answer:

1. **Histogram:** A graphical representation that organizes a group of data points into user-specified ranges. It is essential for understanding the distribution (frequency) of numerical data.
2. **Scatter Plot:** A graph in which the values of two variables are plotted along two axes. It is primarily used to observe relationships or correlations between the two variables (e.g., positive, negative, or no correlation).

5. A retail store removes duplicate customer records collected from multiple branches. Explain what data pre-processing step is used?

Answer: The step used is **Data Cleaning**, specifically **Data Deduplication**.

- When integrating data from multiple branches, the same customer might appear multiple times (e.g., once as "J. Doe" and once as "John Doe").
- Deduplication identifies these redundant entries and merges them into a single, accurate "golden record" to prevent skewed analysis.

6. A dataset contains missing values for income. Discuss how data cleaning can handle this problem?

Answer: Data cleaning can handle missing `Income` values using several imputation techniques:

1. **Ignore** the **tuple:** Remove the row entirely (usually done if the class label is missing, but less effective if the dataset is small).
2. **Fill usually a global constant:** Replace all missing values with "Unknown" or "-1".
3. **Use a central tendency measure:** Replace the missing value with the **mean** or **median** income of the dataset.
4. **Use the most probable value:** Use regression or decision tree inference to predict the likely income.

7. An analyst uses a scatter plot to study the relationship between advertising cost and sales. Explain why this visualization is suitable.

Answer: A scatter plot is suitable because both "Advertising Cost" and "Sales" are **continuous numerical variables**.

- The plot places Cost on the X-axis and Sales on the Y-axis.
- It visually reveals the **correlation**: Does higher spending lead to higher sales? (Positive correlation).
- It also helps identify **outliers** (instances where high spending resulted in low sales, or vice versa).

8. Explain the uses of normalization in a Data Warehouse.

Answer:

- **Data Integrity:** In the **Enterprise Data Warehouse (EDW)** layer (Inmon approach), normalization (3NF) is used to ensure data is atomic and free from redundancy. This maintains a "single version of the truth."
- **Space Efficiency:** It reduces data duplication, which saves storage space in the staging or integration layers.
- **Update Anomaly Prevention:** It ensures that updates happen in only one place, preventing data inconsistencies before the data is moved to denormalized marts.

9. Define OLAP & Explain its role in data warehousing.

Answer: OLAP (Online Analytical Processing) is a computing method that enables users to easily and selectively extract and view data from different points of view.

- **Role:** It allows for multidimensional analysis of data stored in the warehouse.
- It supports complex calculations, trend analysis, and sophisticated modeling (e.g., slice and dice, drill down) that standard SQL queries (OLTP) cannot handle efficiently.

10. Differentiate Meta Data and Data Mart.

Answer:

- **Metadata:** This is "data about data." It describes the structure, content, and context of the data warehouse (e.g., table definitions, data source info, update history). It acts as a directory or catalog.
- **Data Mart:** This is a subset of a data warehouse, usually oriented to a specific business line or team (e.g., Marketing Data Mart, Finance Data Mart). It contains data relevant only to a specific group of users.

11. What are Facts? Define Fact Table.

Answer:

- **Facts:** Facts are the numerical measurements or metrics that a business wants to analyze (e.g., Sales Amount, Quantity Sold, Total Profit).
- **Fact Table:** The central table in a Star Schema that contains these numerical facts and foreign keys referring to the dimension tables. It represents the events or transactions of the business.

12. Explain what Data Warehousing is & its main functions.

Answer: A Data Warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data in support of management's decision-making process.

- **Main Functions:**
    1. **Data Consolidation:** Merges data from heterogeneous sources.
    2. **Historical Storage:** Maintains historical data for trend analysis over time.
    3. **Query Performance:** Optimized for read-heavy analytical queries (OLAP) rather than transaction processing.

13. Discuss the ETL process in Data Warehousing.

Answer: ETL stands for **Extract, Transform, and Load**. It is the backbone of data warehousing.

- **Extract:** Reading data from various source systems (SQL databases, XML files, legacy systems).
- **Transform:** Cleaning, filtering, validating, and applying business rules to the data (e.g., calculating profit, standardizing date formats).
- **Load:** Writing the processed data into the target Data Warehouse.

14. Explain the Star Schema in Data Warehousing & describe its structure with an example.

Answer: The Star Schema is the simplest data warehouse schema.

- **Structure:** It consists of one central **Fact Table** surrounded by multiple **Dimension Tables**. The diagram resembles a star. The dimension tables are *denormalized*.
- **Example:** A `Sales` fact table (containing `Revenue`, `Units_Sold`) is connected to dimensions like `Time`, `Product`, `Store`, and `Customer`.

15. What is a Snowflake Schema? Explain its structure with an example.

Answer: The Snowflake Schema is a variation of the Star Schema where dimension tables are normalized.

- **Structure:** Dimension tables are split into additional tables to remove redundancy. The diagram resembles a snowflake.
- **Example:** In a `Product` dimension, instead of storing `Category` directly, the `Product` table links to a `Category` table, which links to a `Department` table. This saves space but makes queries more complex (more joins).

16. List the difference between OLAP & OLTP.

Answer: | Feature | OLTP (Online Transaction Processing) | OLAP (Online Analytical Processing) | | :--- | :--- | :--- | | **Focus** | Day-to-day operations (insert, update, delete). | Decision support and analysis (read-only). | | **Data** | Current, up-to-the-minute data. | Historical, summarized, and integrated data. | | **Users** | Clerks, DBAs, Operational staff. | Managers, Executives, Analysts. | | **Complexity** | Simple, fast, repetitive transactions. | Complex, ad-hoc queries involving aggregations. |

17. Compare & Contrast an operational database with a Data Warehouse.

Answer:

- **Operational DB:** Optimized for writing data. High availability and speed for small transactions are critical. Used to *run* the business (e.g., an ATM system).
- **Data Warehouse:** Optimized for reading data. High volume data storage and query throughput are critical. Used to *analyze* the business.
- **Contrast:** An operational DB uses highly normalized schemas (3NF) to avoid errors, while a Data Warehouse often uses denormalized schemas (Star Schema) to improve query speed.

18. Why is a Data Mart considered cost-effective compared to a Data Warehouse?

Answer:

- **Scope:** A Data Mart focuses on a single department (e.g., HR), whereas a Warehouse covers the entire enterprise.
- **Development Time:** Marts take weeks/months to build; Warehouses take months/years.
- **Resources:** Marts require less storage, less processing power, and fewer personnel to maintain, leading to significantly lower hardware and implementation costs.

19. List out the various OLAP operations.

Answer:

1. **Roll-up (Drill-up):** Aggregating data (e.g., moving from "Daily" sales to "Monthly" sales).
2. **Drill-down:** Breaking data down into details (e.g., moving from "Country" view to "City" view).
3. **Slice:** Selecting a single dimension (e.g., showing only "2023" data).
4. **Dice:** Selecting a sub-cube by specifying ranges on multiple dimensions.
5. **Pivot (Rotate):** Rotating the data axes to view the data from a different perspective.

20. Design a data mart for a specific department (e.g., Marketing). Justify the selection of structure & schema.

Answer:

- **Department:** Marketing.
- **Structure/Schema: Star Schema**.
- **Justification:** Marketing analysts need to run fast, simple queries (e.g., "How did Campaign A perform in Q1?"). A Star Schema is easier for business users to understand and provides the fastest query performance because it minimizes the number of table joins compared to a Snowflake schema.
- **Data Sources:** Sales transaction logs, Customer CRM data, Google Analytics data.

Section B: Objective / One-Liners

Instructions: These questions require concise, specific answers based on the concepts identified.

21. Infer the purpose of Fact & Dimensional tables in a Star Schema. Answer: Fact tables store quantitative data (measures/numbers), while Dimensional tables store qualitative data (context/descriptions like "who," "where," "when").

22. Identify which of the following is a correct ETL activity. Answer: Cleaning and standardizing data before loading.

23. If an OLAP server stores data in a multidimensional cube for faster query performance, what does MOLAP stand for? Answer: Multidimensional Online Analytical Processing.

24. In Star Schema design, which **option best describes what a dimension table contains? Answer:** It defines dimensions, attributes, and hierarchies used to analyze the facts.

25. A company wants to analyze year-wise sales trends for decision making. Which feature best matches the **required system? Answer:** Support for historical data analysis (Time-variant nature).